

Isselmou Oumar

Paris, FR | +33 6 46 88 17 31 | isselmouo4@gmail.com | [LinkedIn](#) | [GitHub](#)

EXPERIENCE

Quantitative Analyst, *Société Générale* – Paris

Jan 2025 – Present

- Building PD model for €142 B corporate book using stochastic control and dynamic programming to integrate macro-factor dynamics and ESG signals; positioned as key R&D initiative within the bank
- Developed real-time ESG sentiment scoring pipeline with sectoral overlays, generating cross-sector dispersion signals for issuer-level risk-premia ranking
- Parallelized Monte-Carlo credit simulation engine in R (RcppParallel), cutting runtime 40 % across the full corporate book

Quantitative Research Intern, *Amundi Technology* – Paris

Apr 2024 – Oct 2024

- Co-authored research on carbon-price shock transmission; calibrated structural + ML models on CDS-implied spreads to identify mispricing across credit curves
- Embedded probabilistic scenario trees in DCF framework to extract market-implied transition probabilities and price climate-policy tail risk
- Built sector-level transition-risk heat-maps (Python) guiding low-carbon tilt portfolio re-weighting

SELECTED PROJECTS

- C#/.NET back-testing engine for systematic equity strategies with transaction-cost modelling and performance attribution
- C++ derivatives pricing library: Monte-Carlo + neural-network calibration, Longstaff-Schwartz American option pricer benchmarked against market quotes
- Cross-asset structured-product platform: C++ pricer, Python hedging/portfolio builder, JS/Node.js GUI
- Random Forest / SVM models forecasting swaption subscription demand from U.S. insurer data (Python/SQL); validated by senior quants at Crédit Agricole CIB
- Kalman Filter pairs-trading strategy (Python) for cointegrated pairs (BTC/ETH, WTI/Brent, Gold/Silver) with dynamic z-score entry/exit

EDUCATION

Diplôme d'Ingénieur – Financial Engineering – Grenoble INP - Ensimag, Grenoble

2022 – 2024

Graduated with Honors

MSc Quantitative Finance – Grenoble INP - IAE, Grenoble

2023 – 2024

Graduated with Honors

B.S. Applied Mathematics & Computer Science – Université Grenoble Alpes, Grenoble

2019 – 2022

Graduated with High Honors

SKILLS

Technical: Python, C++, C#, R, Java, SQL, Bash | Git, Docker, Kubernetes

Quantitative: Derivatives pricing & calibration, Monte-Carlo simulation, Statistical/ML modelling, Factor-based systematic strategies, Portfolio construction & risk attribution, Multi-asset back-testing, Time-series analysis, Credit spread modelling

Languages: English, French, Arabic, Spanish

LEADERSHIP & COMMUNITY

- Youth football coach, ages 8–14
- Member, Mauritanian Engineering Committee
- Founder, mobile app connecting users to off-peak promotions